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# Privacy Analysis of Augmented Reality Glasses

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### Weekly Report 12

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#### What has been done (Previous Week)

- Decryption efforts for Network traffic:
  - Redid the whole mitmproxy setup and configuration for the rooted phone.
  - Initially my problems before the rooted phone was due to SSL pinning. Now, despite being able to see that Objection and Frida were disabling SSL pinning, the app still failed to trust the proxy's certificate, as I still received some TLS handshake errors for domains like (graph.instagram.com and googleapis.com).  
This could be because of:
    - \* Additional certificate checks
    - \* Custom implementations
  - I looked into the code to see if there are some custom 'trustManager' implementations (from the decompiled apk) but when trying to use my custom script to disable it by hooking into the different methods, there were some problems regarding dynamic hooking and some crashes.  
This could be due to:
    - \* An issue with my script logic
    - \* Obfuscation in the apk.
  - Also, if I try to use transparent mode for mitmproxy, with iptables, I do see traffic like if I go on google but the traffic from the app seems to be blocked.
  - I did try to see if I could change the installed certificate to be used as a system-level certificate using Magisk, but that was unsuccessful due to mounting issues and system integrity protections.
- Decryption efforts for Bluetooth traffic:
  - Redid the experiment to generate traffic for the captures by unpairing and repairing the glasses multiple times. I used the btsnoop\_hci.log (the traffic) and the bt\_config.conf (the keys) files to try and decrypt the traffic.
  - Obtained some keys, including the link key (for classic bluetooth), peripheral/local encryption key (for BLE), identity resolving key.
  - Manually added keys to Wireshark preferences since the UI did not support Bluetooth key configuration.
- Updated the code for the filters to fit more into our analysis goals and also improved the code in general so that it is nicer. Also updated the csv file, which contains all the ip addresses and their info of the experiments I have done so far, to be more accurate, especially for the private addresses.

- Added some documentation for the initial challenges though not finished.

### **What will be done next week (and following weeks as some tasks may extend over multiple weeks)**

- Need to continue for the decryption on both the Network and Bluetooth traffic.
- Write documentations on what has been attempted the last week in regards to decrypting both Network and Bluetooth traffic.
- Redo the images and analysis for the Wi-Fi experiments and upload it.
- Finish the documentation for the initial challenges.
- Begin writing documentation for thesis use, especially on unsuccessful approaches to provide some clarity and/or insights.
- Look into general pattern, protocol, endpoint, statistical analysis for the traffic and observe any correlations but make sure to write them down in documentation this time.

### **Meeting**

Had a meeting with Nima on the 13th, where:

- Continued discussion about the errors and problems encountered in decryption of network traffic
- Tried to go through the mitmproxy and certificate bypass process together
- Had a look at the images generated and discussed about improvements including the filters and categorization of the traffic.